Programming for Artificial Intelligence

[Assignment — 01] [Total Marks 40]

Instructions.

- Combine all your work in one folder. The folder must contain only the files as instructed in each question separately.
- Rename the folder as ROLL-NUM_SECTION_NAME (e.g. 221234_Ali) and compress the folder as a zip file. (e.g. 221234_Ali.zip).
- Do not submit .rar file.
- Submit the .zip file on Google Classroom within the deadline.
- Submission other than Google classroom (e.g. Email etc.) Will not be accepted.
- The student is solely responsible to check the final zip files for issues like corrupt file, virus in the file, mistakenly exe sent.
- If instructor cannot download the file from Google classroom due to any reason it will lead to zero marks in the assignment.
- Deadline to submit assignment is 28th February 2026 11:59 PM.
- Assignment submitted after the deadline will be marked DIRECT ZERO.
- Correct and timely submission of the assignment is the responsibility of every student; hence no relaxation will be given to anyone.
- For timely completion of the assignment, start as early as possible.
- Plagiarism is not allowed. If found plagiarized, you will be awarded zero marks in all the assignments.

Example solution for question 1 to 3

Example Question: Solve the expression: ((8 - 3) * 5 / 2 + 7 > 20) and true or false

Solution:

((8 - 3) * 5 / 2 +7 > 20) and true or false
= ((5 * 5 / 2 +7 > 20) and true) or false
= ((25 / 2 +7 > 20) and true) or false
= ((12.5 + 7 > 20) and true) or false
= (19.5 > 20 and true) or false
= (false or false)
= false

Question 1: [Marks 5]

Solve the expression step by step.

Expression: ((4 + 4) * 3 - 9) / (8 % 3) > 1 and (True or False) and not (5 == 6)

Question 1 Submission Format:

You have to solve it on paper and write your name, roll number, and question number on top of that paper and then take a picture of that solution and submit that picture (format of picture must be jpeg or png. (Picture must be readable, zero marks if there is a single ambiguity in reading the picture). Question1_Rollnumber.jpeg file. For example (Question1_221234.png or Question1_221234.jpeg)

Question 2: [Marks 5]
Solve the expression step by step.

9 / 2 * 7 + 10 % 3 < 27 or (false and true)

Question 2 Submission Format:
You have to solve it on paper and write your name, roll number, and question number on top of that paper and then take a picture of that solution and submit that picture (format of picture must be jpeg or png. (Picture must be readable, zero marks if there is a single ambiguity in reading the picture). Question2_RolInumber.jpeg file. For example (Question2_221234.png or Question2_221234.jpeg)

Question 3: [Marks 5]
Solve the expression step by step.

(2 + 5 * 3 - 7) >= 8 or (true and false) and not(3 > 5)

Question 3 Submission Format:
You have to solve it on paper and write your name, roll number, and question number on top of that paper and then take a picture of that solution and submit that picture (format of picture must be jpeg or png. (Picture must be readable, zero marks if there is a single ambiguity in reading the picture). Question3_RolInumber.jpeg file. For example (Question3_221234.png or Question3_221234.jpeg)

Question 4: [Marks 10]
Write a Python program that:

1. Prints your name, roll number, degree name, and the phrase "Smile its Sunnah" at the start.

Condition: You cannot use any string literal in print() that contains white spaces (so no "Smile its Sunnah" directly).

2. Prints the following smiley pattern using string formatting/alignment techniques (str.center(), format(), or f-strings). **Condition**: You are not allowed to simply hardcode all spaces.

For example:

Hy. I am Mr./Miss (Name)
My roll number is (Your roll number like: 221-1234)
My Degree is (Your Degree Like: BS(AI), BS(CS)..)

Smile its Sunnah:

```
 _______
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| O  O  |
|   ^   |
| \___/ |
\_______/
```

Question 5: [Marks 5]

Write a program that uses input to prompt a user for their name and then welcomes them. Enter your name:

Chuck

Hello Chuck

Question 6: [Marks 10]

Write a program to prompt for a score between 0.0 and 1.0. If the score is out of range, print an error message. If the score is between 0.0 and 1.0, print a grade using the following table:

```
 Score   Grade
>= 0.9     A
>= 0.8     B
>= 0.7     C
>= 0.6     D
 < 0.6     F

Enter score: 0.95
A

Enter score: perfect
Bad score

Enter score: 10.0
Bad score

Enter score: 0.75
C

Enter score: 0.5
F
```